

Stochastic Gradient Descent

The problem of variance

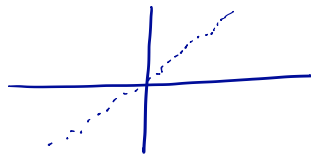
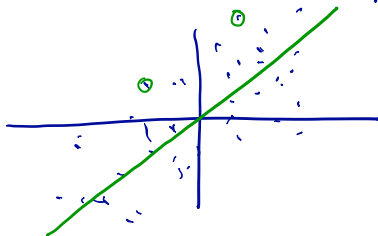
Setting: $f(x) = \frac{1}{n} \sum f_i(x)$

$\rightarrow$ SGD: $x_{t+1} = x_t - \eta \nabla f_I(x_t), \quad I \sim \text{uni} \{1, \dots, n\}$

⊗ Result: $\mathbb{E} f\left(\frac{1}{T} \sum x_t\right) - f(x^*) \leq \frac{\|x_1 - x^*\|}{\sqrt{T}} \cdot \sqrt{\mathbb{E} [\|\nabla f_I(x)\|_2^2]}$

1
"variance term"

Picture:



Stochastic Gradient Descent - mini batch

Rather than picking a single f_i and using ∇f_i
instead pick B f_i 's : $f_{i_1}, f_{i_2}, \dots, f_{i_B}$,
and ~~use~~ use the average gradient:

Mini-batch SGD:

$$x_{t+1} = x_t - \eta \underbrace{\frac{1}{B} \sum_{j=1}^B \nabla f_{I_j}(x_t)}$$

$I_1, I_2, \dots, I_j \sim \text{uni}\{1, \dots, n\}$
independent

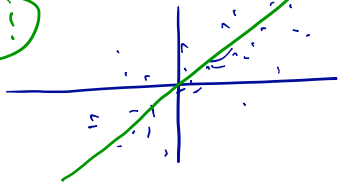
Check:
$$\mathbb{E} \left[\frac{1}{B} \sum_{j=1}^B \nabla f_{I_j}(x_t) \right] = \frac{1}{B} \sum_{j=1}^B \boxed{\mathbb{E} \nabla f_{I_j}(x_t)}$$

$$= \frac{1}{B} \sum_{j=1}^B \nabla f(x_t) = \nabla f(x_t)$$

Stochastic Gradient Descent - mini batch

Does it help?

YES!



Advantages:

1. Reduces variance!
2. Mini-batch is parallelizable.

Batch of size B -
$$\tilde{g}_x = \frac{1}{B} \sum_{j=1}^B \nabla f_{I_j}(x)$$

each of the gradient terms ∇f_{I_j} can be computed independently, i.e. in parallel.

Disadvantages:

1. More work (B times more) per iteration of alg.
2. still cannot exploit self-tuning in case when f is smooth.